# NATIONAL QUANTUM MISSION AND QUANTUM TECHNOLOGY

**BIT AND QUBIT CARD → QUANTUM PRINCIPLE SPLIT → ALGORITHM MECHANISM RAIL → DECOHERENCE ENGINEERING STACK → PHYSICAL-TO-LOGICAL LADDER → FOUR TECHNOLOGY FAMILIES**

Read the thick cyan rail from Stage 00 to the final core synthesis. Each card uses a concept-appropriate structure; the grey E card is optional enrichment only and never repairs missing core content.

◆ PRIMARY CORE FLOW ◈ SUBORDINATE EXTRA ★ APPROVAL / REVIEW

CORE BEFORE EXTRA • prior artefacts unchanged • exact same master drives poster and tiles

Approval: FALSE • Pending user review • source generation g2 and all prior artifacts unchanged

00

## STAGE 00 BIT AND QUBIT CARD

BIT AND QUBIT CARD · CLASSICAL BIT · RECORDED AS 0 OR 1 · AMPLITUDES OVER BASIS STATES · ONE CLASSICAL OUTCOME WITH PROBABILITY · NOT THE DEFINING DIFFERENCE · STATE DESCRIPTION IS NOT TWO STORED CLASSICAL BITS

| SYSTEM / DEVICE / INSTITUTION | DECISIVE TECHNICAL DISTINCTION | STATUS / UNIT / EVIDENCE LIMIT |
|---|---|---|
| CLASSICAL BIT → recorded as 0 or 1 | MEASUREMENT → one classical outcome with probability | RULE → state description is not two stored classical bits |
| QUBIT → amplitudes over basis states | SPEED → not the defining difference | CLASSICAL BIT → recorded as 0 or 1 |

**SYSTEM / DEVICE / INSTITUTION**
- CLASSICAL BIT → recorded as 0 or 1
- QUBIT → amplitudes over basis states

**DECISIVE TECHNICAL DISTINCTION**
- MEASUREMENT → one classical outcome with probability
- SPEED → not the defining difference

**STATUS / UNIT / EVIDENCE LIMIT**
- RULE → state description is not two stored classical bits
- CLASSICAL BIT → recorded as 0 or 1

**ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM: RULE → state description is not two stored classical bits.**

01

## STAGE 01 QUANTUM PRINCIPLE SPLIT

QUANTUM PRINCIPLE SPLIT · ONE SYSTEM IN A COMBINATION OF BASIS STATES · JOINT STATE OF TWO OR MORE SYSTEMS · ARBITRARY UNKNOWN STATE CANNOT BE COPIED PERFECTLY · NO SIGNALLING · ENTANGLEMENT DOES NOT TRANSMIT MESSAGES FASTER · EACH PRINCIPLE ANSWERS A DIFFERENT QUESTION

| SYSTEM / DEVICE / INSTITUTION | DECISIVE TECHNICAL DISTINCTION | STATUS / UNIT / EVIDENCE LIMIT |
|---|---|---|
| SUPERPOSITION → one system in a combination of basis states | NO-CLONING → arbitrary unknown state cannot be copied perfectly | RULE → each principle answers a different question |
| ENTANGLEMENT → joint state of two or more systems | NO SIGNALLING → entanglement does not transmit messages faster | SUPERPOSITION → one system in a combination of basis states |

**SYSTEM / DEVICE / INSTITUTION**
- SUPERPOSITION → one system in a combination of basis states
- ENTANGLEMENT → joint state of two or more systems

**DECISIVE TECHNICAL DISTINCTION**
- NO-CLONING → arbitrary unknown state cannot be copied perfectly
- NO SIGNALLING → entanglement does not transmit messages faster

**STATUS / UNIT / EVIDENCE LIMIT**
- RULE → each principle answers a different question
- SUPERPOSITION → one system in a combination of basis states

**ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM: RULE → each principle answers a different question.**

02

## STAGE 02 ALGORITHM MECHANISM RAIL

ALGORITHM MECHANISM RAIL · PREPARE STATE · INITIALISE QUBITS · QUANTUM GATES · TRANSFORM AMPLITUDES · SUPPRESS AND REINFORCE PATHS · OBTAIN CLASSICAL SAMPLE · SUPERPOSITION DOES NOT REVEAL EVERY ANSWER

PREPARE STATE → initialise qubits → QUANTUM GATES → transform amplitudes → INTERFERENCE → suppress and reinforce paths → MEASUREMENT → obtain classical sample → TRAP → superposition does not reveal every answer

**STARTING CONDITION / INPUT**
- PREPARE STATE → initialise qubits
- QUANTUM GATES → transform amplitudes

**SCIENTIFIC OR ENGINEERING MECHANISM**
- INTERFERENCE → suppress and reinforce paths
- MEASUREMENT → obtain classical sample

**OUTPUT / FAILURE BOUNDARY**
- TRAP → superposition does not reveal every answer
- PREPARE STATE → initialise qubits

**ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM: TRAP → superposition does not reveal every answer.**

03

## STAGE 03 DECOHERENCE ENGINEERING STACK

DECOHERENCE ENGINEERING STACK · ENVIRONMENTAL INTERACTION · PHASE INFORMATION DEGRADES · COHERENCE TIME · FINITE COMPUTATION WINDOW · REDUCE DISTURBANCE · CONTROL ELECTRONICS · EXECUTE ACCURATE GATES

**STARTING CONDITION / INPUT**
- ENVIRONMENTAL INTERACTION → phase information degrades
- COHERENCE TIME → finite computation window

**SCIENTIFIC OR ENGINEERING MECHANISM**
- CRYOGENICS / VACUUM / SHIELDING → reduce disturbance
- CONTROL ELECTRONICS / MATERIALS → execute accurate gates

**OUTPUT / FAILURE BOUNDARY**
- OUTCOME → engineering, not theory alone, limits scale
- ENVIRONMENTAL INTERACTION → phase information degrades

**ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM: OUTCOME → engineering, not theory alone, limits scale.**

04

## STAGE 04 PHYSICAL-TO-LOGICAL LADDER

PHYSICAL-TO-LOGICAL LADDER · PHYSICAL QUBIT · NOISY HARDWARE ELEMENT · ERROR CHARACTERISATION · MEASURE FAILURE CHANNELS · ERROR-CORRECTING CODE · REDUNDANT ENCODING · LOGICAL QUBIT

PHYSICAL QUBIT → noisy hardware element

ERROR CHARACTERISATION → measure failure channels → ERROR-CORRECTING CODE → redundant encoding → LOGICAL QUBIT → protected information unit

**STARTING CONDITION / INPUT**
- PHYSICAL QUBIT → noisy hardware element
- ERROR CHARACTERISATION → measure failure channels

**SCIENTIFIC OR ENGINEERING MECHANISM**
- ERROR-CORRECTING CODE → redundant encoding
- LOGICAL QUBIT → protected information unit

**OUTPUT / FAILURE BOUNDARY**
- RULE → raw qubit count is not useful capability
- PHYSICAL QUBIT → noisy hardware element

**ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM: RULE → raw qubit count is not useful capability.**

05

## STAGE 05 FOUR TECHNOLOGY FAMILIES

FOUR TECHNOLOGY FAMILIES · SELECTED INFORMATION-PROCESSING ALGORITHMS · QUANTUM-STATE LINKS AND KEY DISTRIBUTION · PRECISION MEASUREMENT · ENABLING HARDWARE SUBSTRATE · QUANTUM TECHNOLOGY IS NOT COMPUTING ALONE

| SYSTEM / DEVICE / INSTITUTION | DECISIVE TECHNICAL DISTINCTION | STATUS / UNIT / EVIDENCE LIMIT |
|---|---|---|
| COMPUTING → selected information-processing algorithms | SENSING / METROLOGY → precision measurement | TRAP → quantum technology is not computing alone |
| COMMUNICATION → quantum-state links and key distribution | MATERIALS / DEVICES → enabling hardware substrate | COMPUTING → selected information-processing algorithms |

**SYSTEM / DEVICE / INSTITUTION**
- COMPUTING → selected information-processing algorithms
- COMMUNICATION → quantum-state links and key distribution

**DECISIVE TECHNICAL DISTINCTION**
- SENSING / METROLOGY → precision measurement
- MATERIALS / DEVICES → enabling hardware substrate

**STATUS / UNIT / EVIDENCE LIMIT**
- TRAP → quantum technology is not computing alone
- COMPUTING → selected information-processing algorithms

**ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM: TRAP → quantum technology is not computing alone.**

06

## STAGE 06 QKD SECURITY CHAIN

QKD SECURITY CHAIN · QUANTUM STATES · DISTRIBUTE KEY MATERIAL · DISTURBANCE TEST · REVEAL POSSIBLE INTERCEPTION · AUTHENTICATED ENDPOINTS · PREVENT IMPERSONATION · CLASSICAL ENCRYPTION

QUANTUM STATES → distribute key material → DISTURBANCE TEST → reveal possible interception → AUTHENTICATED ENDPOINTS → prevent impersonation → CLASSICAL ENCRYPTION → protects message content → RULE → QKD alone is not a complete secure network

**STARTING CONDITION / INPUT**
- QUANTUM STATES → distribute key material
- DISTURBANCE TEST → reveal possible interception

**SCIENTIFIC OR ENGINEERING MECHANISM**
- AUTHENTICATED ENDPOINTS → prevent impersonation
- CLASSICAL ENCRYPTION → protects message content

**OUTPUT / FAILURE BOUNDARY**
- RULE → QKD alone is not a complete secure network
- QUANTUM STATES → distribute key material

**ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM: RULE → QKD alone is not a complete secure network.**

07

## STAGE 07 QKD VERSUS PQC

QKD VERSUS PQC · QUANTUM HARDWARE AND PHYSICAL LINK · CLASSICAL QUANTUM-RESISTANT ALGORITHMS · QKD PURPOSE · SPECIALISED KEY DISTRIBUTION · PQC PURPOSE · BROAD STANDARDS AND SOFTWARE MIGRATION · COMPLEMENTARY TOOLS WITH DIFFERENT TIMELINES

| SYSTEM / DEVICE / INSTITUTION | DECISIVE TECHNICAL DISTINCTION | STATUS / UNIT / EVIDENCE LIMIT |
|---|---|---|
| QKD → quantum hardware and physical link | QKD PURPOSE → specialised key distribution | POLICY → complementary tools with different timelines |
| PQC → classical quantum-resistant algorithms | PQC PURPOSE → broad standards and software migration | QKD → quantum hardware and physical link |

**SYSTEM / DEVICE / INSTITUTION**
- QKD → quantum hardware and physical link
- PQC → classical quantum-resistant algorithms

**DECISIVE TECHNICAL DISTINCTION**
- QKD PURPOSE → specialised key distribution
- PQC PURPOSE → broad standards and software migration

**STATUS / UNIT / EVIDENCE LIMIT**
- POLICY → complementary tools with different timelines
- QKD → quantum hardware and physical link

**ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM: PQC PURPOSE → broad standards and software migration.**

08

## STAGE 08 NQM INSTITUTION MAP

NQM INSTITUTION MAP · MINISTRY OF SCIENCE AND TECHNOLOGY · MINISTERIAL UMBRELLA · IMPLEMENTING DEPARTMENT · NATIONAL CAPABILITY MISSION · FOUR VERTICAL ECOSYSTEM ANCHORS · DEPARTMENT, MISSION AND HUB ARE DISTINCT

MINISTRY OF SCIENCE AND TECHNOLOGY → ministerial umbrella

DST → implementing department → NQM → national capability mission → T-HUBS → four vertical ecosystem anchors

**CONCEPT / ARCHITECTURE**
- MINISTRY OF SCIENCE AND TECHNOLOGY → ministerial umbrella
- DST → implementing department

**MECHANISM / APPLICATION**
- NQM → national capability mission
- T-HUBS → four vertical ecosystem anchors

**READINESS / STATUS LIMIT**
- RULE → department, mission and hub are distinct
- MINISTRY OF SCIENCE AND TECHNOLOGY → ministerial umbrella

**ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM: RULE → department, mission and hub are distinct.**

09

## STAGE 09 FOUR T-HUB MAP

FOUR T-HUB MAP · IISC BENGALURU · QUANTUM COMPUTING · IIT MADRAS WITH C-DOT · QUANTUM COMMUNICATION · IIT BOMBAY · QUANTUM SENSING AND METROLOGY · IIT DELHI

| SYSTEM / DEVICE / INSTITUTION | DECISIVE TECHNICAL DISTINCTION | STATUS / UNIT / EVIDENCE LIMIT |
|---|---|---|
| IISc BENGALURU → Quantum Computing | IIT BOMBAY → Quantum Sensing and Metrology | TRAP → hub creation is not operational capability |
| IIT MADRAS WITH C-DOT → Quantum Communication | IIT DELHI → Quantum Materials and Devices | MUST REMEMBER: Quantum technology covers computing, communication and sensing: qubits... |

**SYSTEM / DEVICE / INSTITUTION**
- IISc BENGALURU → Quantum Computing
- IIT MADRAS WITH C-DOT → Quantum Communication

**DECISIVE TECHNICAL DISTINCTION**
- IIT BOMBAY → Quantum Sensing and Metrology
- IIT DELHI → Quantum Materials and Devices

**STATUS / UNIT / EVIDENCE LIMIT**
- TRAP → hub creation is not operational capability
- MUST REMEMBER: Quantum technology covers computing, communication and sensing: qubits...

**ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM: TRAP → hub creation is not operational capability.**

10

## STAGE 10 READINESS LADDER

READINESS LADDER · CABINET APPROVAL · MISSION AUTHORITY AND OUTLAY · INTENDED FUTURE CAPABILITY · RESEARCH PROTOTYPE · BOUNDED DEMONSTRATION · VALIDATED SYSTEM · REPEATABLE PERFORMANCE EVIDENCE

CABINET APPROVAL → mission authority and outlay

TARGET → intended future capability → RESEARCH PROTOTYPE → bounded demonstration → VALIDATED SYSTEM → repeatable performance evidence

**STARTING CONDITION / INPUT**
- CABINET APPROVAL → mission authority and outlay
- TARGET → intended future capability

**SCIENTIFIC OR ENGINEERING MECHANISM**
- RESEARCH PROTOTYPE → bounded demonstration
- VALIDATED SYSTEM → repeatable performance evidence

**OUTPUT / FAILURE BOUNDARY**
- OPERATIONAL DEPLOYMENT → field use with sustainment
- CLOSE DISTINCTION: Qubit is not a faster classical bit, gate count is not useful...

**ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM: CLOSE DISTINCTION: Qubit is not a faster classical bit, gate count is not useful.**

11

## STAGE 11 QUANTUM CLAIM FIREWALL

QUANTUM CLAIM FIREWALL · QUBIT, SUPERPOSITION, ENTANGLEMENT AND DECOHERENCE · COMPUTING, COMMUNICATION, SENSING OR MATERIALS · PHYSICAL AND LOGICAL QUBITS; QKD AND PQC · APPROVAL, TARGET, PROTOTYPE, VALIDATION AND OPERATION · OUTLAY, QUBITS, DISTANCE AND STARTUP CLAIMS OFFICIALLY · EVIDENCE LIMIT · IDENTIFY PLATFORM, QUBIT/MEASUREMENT

DEFINE → qubit, superposition, entanglement and decoherence → CLASSIFY → computing, communication, sensing or materials → SEPARATE → physical and logical qubits; QKD and PQC → LABEL → approval, target, prototype, validation and operation → VERIFY → outlay, qubits, distance and startup claims officially

**SYSTEM / DEVICE / INSTITUTION**
- DEFINE → qubit, superposition, entanglement and decoherence
- CLASSIFY → computing, communication, sensing or materials

**DECISIVE TECHNICAL DISTINCTION**
- SEPARATE → physical and logical qubits; QKD and PQC
- LABEL → approval, target, prototype, validation and operation

**STATUS / UNIT / EVIDENCE LIMIT**
- VERIFY → outlay, qubits, distance and startup claims officially
- EVIDENCE LIMIT: MECHANISM / STATUS / CAUSATION: Identify platform, qubit/measurement...

MECHANISM / VERDICT — EVIDENCE LIMIT: MECHANISM / STATUS / CAUSATION: Identify platform, qubit/measurement...

**ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM: EVIDENCE LIMIT: MECHANISM / STATUS / CAUSATION: Identify platform, qubit/measurement.**

E

## EXTRA SUBORDINATE ENRICHMENT — ONLY AFTER THE COMPLETE CORE

OPTIONAL ADVANCED ENRICHMENT · QUANTUM CAPABILITY IS PATH-DEPENDENT · NATIONS THAT BUILD TALENT, FABRICATION KNOW-HOW AND STANDARDS EARLY · THE MOST BINDING CONSTRAINTS ARE OFTEN ENGINEERING AND ECOSYSTEM · A MISSION APPROACH IS JUSTIFIED BECAUSE FRAGMENTED PROJECTS ALONE · NATIONAL MISSION FRAMEWORK · T-HUB AT IISC BENGALURU · QUANTUM COMPUTING.

**OPTIONAL SCIENTIFIC / ENGINEERING DEPTH**
- CAUTION: Quantum capability is path-dependent: nations that build talent, fabrication know-how and standards early can gain long-term advantages.
- CAUTION: The most binding constraints are often engineering and ecosystem constraints rather than pure theory.
- CAUTION: A mission approach is justified because fragmented projects alone may not create platform-level capability.
- NQM: national mission framework.

**READINESS, UNIT OR STATUS LIMIT**
- T-Hub at IISc Bengaluru: Quantum Computing.
- T-Hub at IIT Madras with C-DOT: Quantum Communication.
- T-Hub at IIT Bombay: Quantum Sensing & Metrology.
- T-Hub at IIT Delhi: Quantum Materials & Devices.

**COMPARATIVE TECHNOLOGY / GOVERNANCE NUANCE**
- PIB parliamentary answer: says the T-Hubs comprise 14 Technical Groups across 17 States and 2 UTs.
- Quantum communication is strategically relevant for secure networks and national-security communication contexts.
- Quantum sensing/metrology can matter for navigation, clocks, precision instrumentation and advanced measurement ecosystems.
- Quantum materials/devices are foundational because hardware dependence can otherwise cripple mission ambition.

OPTIONAL STATUS — This optional depth is unnecessary for a competent core answer; use it only for added nuance after the complete core has been written.

END OF MASTER CANVAS • poster embeds this exact master • tiled pages are overlapping crops of the same pixels